

■ 3.5.2 Total Derivatives

<code>Dt[f]</code>	total differential df
<code>Dt[f, x]</code>	total derivative $\frac{df}{dx}$
<code>Dt[f, x₁, x₂, ...]</code>	multiple total derivative $\frac{d}{dx_1} \frac{d}{dx_2} \dots f$
<code>Dt[f, x, Constants -> {c₁, c₂, ...}]</code>	total derivative with c_i constant (i.e. $dc_i = 0$)
<code>y/: Dt[y, x] = 0</code>	set $\frac{dy}{dx} = 0$
<code>SetAttributes[c, Constant]</code>	define c to be a constant in all cases

Total differentiation operations.

When you find the derivative of some expression f with respect to x , you are effectively finding out how fast f changes as you vary x . Often f will depend not only on x , but also on other variables, say y and z . The results that you get then depend on how you assume that y and z vary as you change x .

There are two common cases. Either y and z are assumed to stay fixed when x changes, or they are allowed to vary with x . In a standard *partial derivative* $\frac{\partial f}{\partial x}$, all variables other than x are assumed fixed. On the other hand, in the *total derivative* $\frac{df}{dx}$, all variables are allowed to change with x .

In *Mathematica*, `D[f, x]` gives a partial derivative, with all other variables assumed independent of x . `Dt[f, x]` gives a total derivative, in which all variables are assumed to depend on x . In both cases, you can add an argument to give more information on dependencies.

This gives the *partial derivative* $\frac{\partial}{\partial x}(x^2 + y^2)$.
 y is assumed to be independent of x .

```
In[1]:= D[x^2 + y^2, x]
Out[1]= 2 x
```

This gives the *total derivative* $\frac{d}{dx}(x^2 + y^2)$.
 Now y is assumed to depend on x .

```
In[2]:= Dt[x^2 + y^2, x]
Out[2]= 2 x + 2 y Dt[y, x]
```

You can make a replacement for $\frac{dy}{dx}$.

```
In[3]:= % /. Dt[y, x] -> yx
Out[3]= 2 x + 2 y yx
```

You can also make an explicit definition for $\frac{dy}{dx}$. You need to use `y/:` to make sure that the definition is associated with y .

```
In[4]:= y/: Dt[y, x] = 0
Out[4]= 0
```

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With this definition made, `Dt` treats `y` as independent of `x`.

```
In[5]:= Dt[x^2 + y^2 + z^2, x]
Out[5]= 2 x + 2 z Dt[z, x]
```

This removes your definition for the derivative of `y`.

```
In[6]:= Clear[y]
```

This takes the total derivative, with `z` held fixed.

```
In[7]:= Dt[x^2 + y^2 + z^2, x, Constants->{z}]
Out[7]= 2 x + 2 y Dt[y, x]
```

This specifies that `c` is a constant under differentiation.

```
In[8]:= SetAttributes[c, Constant]
```

The variable `c` is taken as a constant.

```
In[9]:= Dt[a^2 + c x^2, x]
Out[9]= 2 c x + 2 a Dt[a, x]
```

The *function* `c` is also assumed to be a constant.

```
In[10]:= Dt[a^2 + c[x] x^2, x]
Out[10]= 2 a Dt[a, x] + 2 x c[x]
```

This gives the total differential $d(x^2 + cy^2)$.

```
In[11]:= Dt[x^2 + c y^2]
Out[11]= 2 x Dt[x] + 2 c y Dt[y]
```

You can make replacements and assignments for total differentials.

```
In[12]:= % /. Dt[y] -> dy
Out[12]= 2 c dy y + 2 x Dt[x]
```